

Faculty Research Grant & Publication Tracker

Complete Requirements Table — Problem Statement #09

Functional Requirements

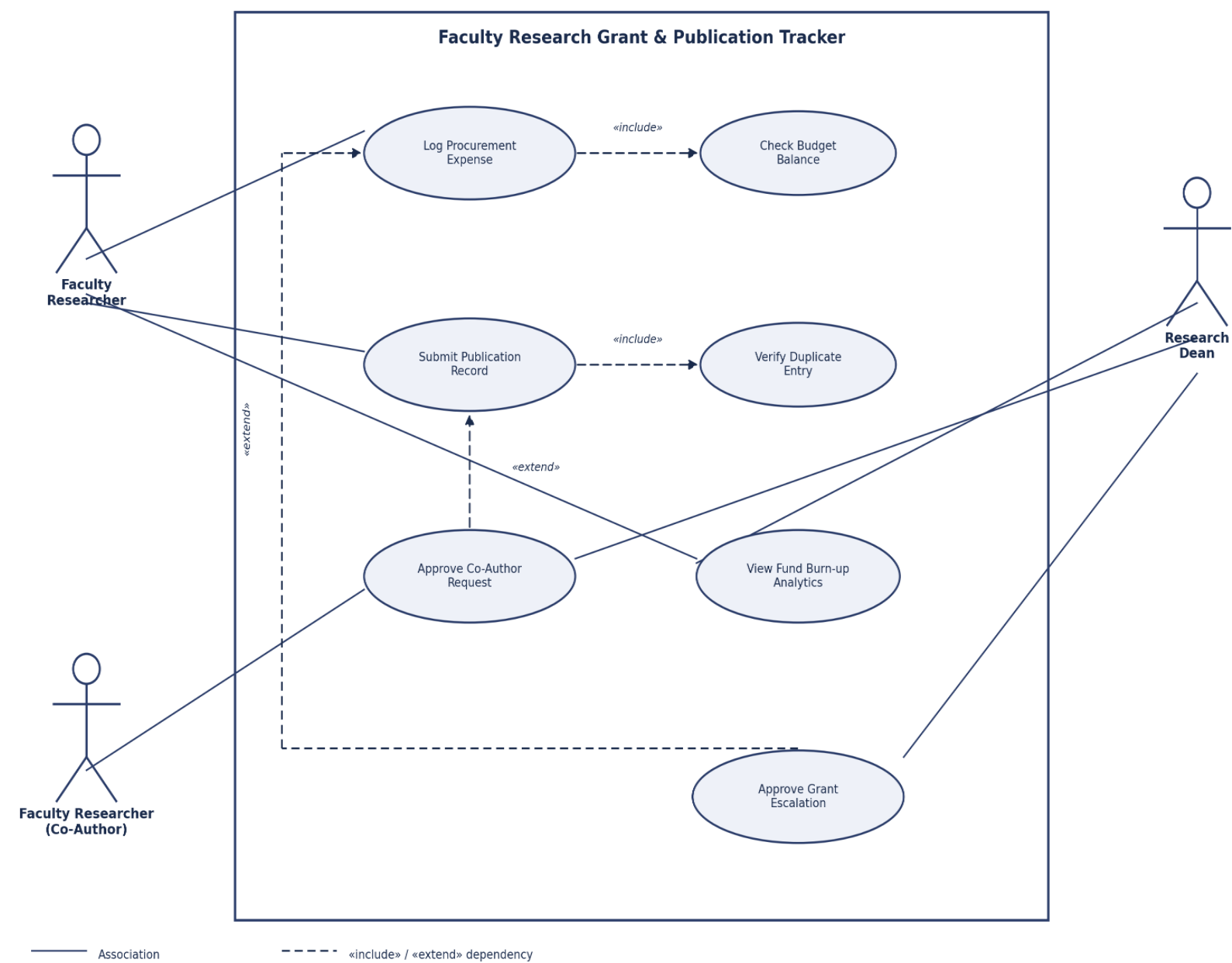
ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall track grant fund allocations, allow faculty to log equipment/consumable procurement expenses against a grant, and compute the remaining budgetary balance in real time.	High	Pass: Balance updates accurately after each expense and the system blocks any expense that would exceed the grant cap unless an admin override is applied. Fail: Overdraft is allowed without an admin override, or the displayed balance does not match the sum of allocations minus expenses.	Prevents grant overdrafts and gives faculty real-time visibility into spendable funds, which is the core financial control the research deanery needs.
FR-002	The system shall allow faculty to submit publication records including title, authors, venue, DOI, indexing status (Scopus/WoS/UGC-CARE), and citation count, and shall check each submission for duplicate DOI/title entries before acceptance.	High	Pass: A duplicate DOI or high-similarity title is detected and the submission is blocked with a clear conflict message; unique submissions are stored successfully. Fail: A duplicate record is silently created, or a valid unique submission is rejected.	Accurate, de-duplicated publication data is essential for correct citation/indexing metrics reported to the deanery and external ranking bodies.
FR-003	For publications with more than one author, the system shall route a co-author approval request to each declared co-author and shall only mark the publication as “verified” once every co-author has approved the authorship order and contribution share.	Medium	Pass: Publication status changes to “verified” only after 100% of co-authors approve; a single rejection reverts the record to draft. Fail: A publication is marked verified while a co-author approval is still pending or rejected.	Protects academic integrity and ensures all contributing authors have consented to how credit is recorded before it counts toward departmental metrics.
FR-004	The system shall generate fund burn-up analytics, visually plotting cumulative grant expenditure against sanctioned budget and time-based milestones, filterable by grant, category, and date range.	Medium	Pass: The burn-up chart renders correct cumulative totals matching the underlying ledger for any selected filter combination. Fail: Chart totals diverge from the ledger, or filters produce no data / incorrect date ranges.	Gives the Research Dean an at-a-glance view of spending velocity across grants, supporting proactive budget management ahead of grant closure deadlines.
FR-005	The system shall automatically notify the Research Dean whenever a grant's cumulative fund utilization crosses the 75%, 90%, or 100% threshold, and shall require Dean approval before further expenses can be logged past the breached threshold.	High	Pass: A notification is generated within the same transaction that crosses a threshold, and further expense logging is blocked until the Dean approves or rejects escalation. Fail: Expenses continue to be logged past a threshold with no notification or approval gate.	Provides an early-warning and control mechanism so grants are not exhausted or overdrawn without deanery visibility and sign-off.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Security & Auditability	The system shall maintain an immutable, tamper-evident audit ledger for all financial approvals (expense logging, escalation decisions) and journal/publication status modifications, recording actor, timestamp, and before/after state.	High	Pass: Benchmarking and penetration tests confirm every financial/status-change event produces a cryptographically verifiable ledger entry that cannot be edited or deleted, under simulated peak load. Fail: Any ledger entry can be silently altered, or an approval/status change occurs without a corresponding entry.	Sponsored research funds and academic records require a defensible, tamper-proof trail for internal audits and external funding-agency compliance reviews.

NFR-002	Availability & Scalability	The system shall be available with at least 99.5% uptime during active grant-submission and publication-reporting windows, and shall support at least 500 concurrent authenticated users without degradation of response time beyond 2 seconds for standard read operations.	Medium	Pass: Load testing at 500 concurrent users maintains sub-2-second response for standard reads and uptime monitoring shows ≥99.5% availability over a rolling 30-day window. Fail: Response times exceed target under load, or uptime falls below 99.5% during a reporting window.	The portal must remain responsive during peak periods such as grant-renewal deadlines and annual publication reporting cycles, when faculty and deanery usage spikes simultaneously.
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UML USE CASE DIAGRAM:



Use Case ID: UC-01

Primary Actor(s): Faculty Researcher

Use Case Name: Log Procurement Expense

Secondary Actor(s): System (Budget Engine)

Brief Description

Allows a Faculty Researcher to record an equipment or consumable procurement expense against a sponsored grant. The system always performs a budget-balance check (an included use case) before the expense is committed, ensuring the grant is never overdrawn without explicit administrative override.

Preconditions

- The Faculty Researcher is authenticated and is a listed Principal/Co-Investigator on an active grant.
- The grant record exists in the system with a defined sanctioned budget and category-wise allocation.

Postconditions

- The expense entry is persisted with a timestamp, category, and supporting invoice reference.
- The grant's remaining budgetary balance is recalculated and updated.
- An immutable audit-ledger entry is created for the transaction (per NFR-001).

Main Success Scenario

1. Faculty Researcher selects an active grant and chooses “Log Procurement Expense.”
2. System displays the current allocation, category-wise spend, and remaining balance.
3. Faculty Researcher enters expense details: amount, category, vendor, and invoice/receipt reference.
4. System «includes» Check Budget Balance to validate the expense against the remaining budget and category cap.
5. Check Budget Balance confirms sufficient funds are available and returns control to the main flow.
6. System commits the expense, updates the remaining balance, and writes an audit-ledger entry.

7. System displays a confirmation with the updated balance to the Faculty Researcher.

Alternate Flow — Expense Exceeds Remaining Balance

1. At step 4, Check Budget Balance determines the requested expense would exceed the remaining budget or category cap.
2. System blocks the transaction and displays a warning indicating the shortfall amount.
3. System offers the Faculty Researcher the option to request an administrative override from the Research Dean.
4. If an override is requested, the system routes a notification to the Research Dean and the use case ends pending approval.
5. If no override is requested, the Faculty Researcher may revise the expense amount and resume at step 3.

Exception / Error Flow

If the invoice reference is missing or malformed, the system rejects the entry at step 3 and prompts the Faculty Researcher to supply a valid reference before proceeding. If the grant record is locked (e.g., grant closure in progress), the system aborts the use case and displays a closure notice.

Notes / UML Relationship Detail

Check Budget Balance is modeled as a separate use case that is mandatorily invoked via an «include» relationship from Log Procurement Expense, since balance validation is required on every execution regardless of path taken and represents reusable behavior shared with other budget-affecting use cases.